Kingdom of Saudi Arabia
Ministry of Education
King Faisal University
College of Computer Sciences & Information Technology

NajmTech: Enhanced Accident Procedures and Motorist Services in KSA.

A project submitted
In partial fulfillment of the requirements for Advanced Software Engineering CS411.

by

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March, 2026

ACKNOWLEDGEMENT

We are deeply grateful for the invaluable guidance and support provided by Dr. Noor Hafsa throughout the Advanced Software Engineering (CS411) course. Her expertise and mentorship were instrumental in the successful execution of this project, "NajmTech: Enhanced Accident Procedures and Motorist Services in KSA." Furthermore, we thank our dedicated group members Hissah Almuhaysh, Latifah Alhafith, Rana Althafar, and Ftoon Althafar for their commitment and contributions across all phases of development, from abstract generation to modeling, testing, and final documentation.

UNDERTAKING

This is to declare that the project entitled "NajmTech: Enhanced Accident Procedures and Motorist Services in KSA" is an original work done by the undersigned, in *partial fulfillment of the requirements for Advanced Software Engineering CS411*, College of Computer Sciences and Information Technology, King Faisal University.

All the analysis, design, and system development have been accomplished by the undersigned. Moreover, this project has not been submitted to any other college or university.

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ABSTRACT

NajmTech is an initiative-based application created in the Kingdom of Saudi Arabia that is strategically inclined towards redefining the process of accidents and improving motorist services using advanced software engineering applications. Fully aligned with the objectives of Saudi Arabia's Vision 2030, NajmTech transforms traditional approaches to accident management through the implementation of innovative technologies, particularly autonomous drones. It offers essential features, and users can log in safely and obtain access to incident history and communication channels with personalized access. Most importantly, a drone can be requested in the event of an accident, where another drone will take real-time pictures and clear shots of the area to help the emergency responders arrive at the scene faster. The system is transparent in the sense that it keeps the users informed on the status of their request through a strong notification system. With ongoing enhancements and cooperation, NajmTech can achieve new levels of excellence in accident management and driver support throughout the Kingdom, thereby making roads much safer for all.

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1. Introduction

In an era marked by rapid technological advancements and evolving transportation systems, the necessity for innovative solutions to enhance road safety and motorist services has become increasingly evident. Traffic accidents represent one of the main causes of human mortality, with the percentage of global traffic accidents increasing to become the third expected cause of death worldwide in 2020 [1]. The Kingdom of Saudi Arabia, like many nations, faces challenges in managing accidents and providing timely assistance to motorists. This challenging environment necessitates a paradigm shift toward more efficient and effective accident procedures, which is precisely the goal of the NajmTech project, poised to revolutionize accident management and elevate the standard of motorist aid in the Kingdom. The project leverages the latest advancements in technology and software engineering to streamline accident procedures and enhance overall driving experience [2].

1.1 Background and Context

The rate of road traffic injuries in the Eastern Mediterranean Region is assessed by the World Health Organization (WHO) to be the second highest in the world. In Saudi Arabia, over 460,000 car accidents occur every year, a number that rises significantly during busy periods such as Ramadan and the Hajj season [1]. This situation emphasizes the urgency for smart solutions to protect the workforce and achieve national safety goals. NajmTech is committed to redefining accident procedures and enhancing motorist services through cutting-edge software engineering solutions. This initiative is fully aligned with the goals of Saudi Arabia's Vision 2030, which emphasizes innovation and sustainable growth. NajmTech utilizes advanced technology to transform the Kingdom's road safety and motorist assistance environments.

1.2 Technological Innovation and Methodology

The core methodology of NajmTech involves integrating the most recent technological developments with software engineering solutions to optimize accident management processes. A key component of this approach is the strategic inclusion of drones (Unmanned Aerial Vehicles, or UAVs), utilized for data collection and real-time aerial monitoring of accident scenes. Drones

are considered a superior method to human presence in detection because of their fast mobility, cost-effectiveness, and ease of implementation [1]. By leveraging drone technology, NajmTech aims to significantly improve first responders' situational awareness, accelerate accident reaction times, and enhance the overall effectiveness of accident management procedures. The drone will capture real-time images and provide clear footage of the accident scene directly to emergency responders. This approach contributes to creating a safer and more resilient transportation environment while facilitating quicker and more efficient responses to incidents.

1.3 Problem Statement

The failure to realize next-generation accident management stems from the current system's reliance on manual data collection, which is time-intensive and leads to poor data quality [3]. The existing process, the as-is model, suffers from several operational issues due to dependency on a Najm worker's physical presence at the scene. This current process causes a long response time due to the delay in the worker's arrival at the accident destination, particularly on crowded roads. The simulation demonstrated that the "Arrive at the Destination" activity is a bottleneck of the overall operation, consuming 20 minutes on average and totaling 20,000 minutes out of the full process time for 1000 instances [1]. Additionally, traditional paper-based documentation is classified as a Non-Value Added (NVA) activity, is prone to errors (e.g., lost paper or poor image quality), and compromises the speed and accuracy of assessment [4].

1.4 Objectives

The proposed NajmTech project aims to resolve these issues by creating a single, highly integrated, anticipatory solution that integrates drone vision and automated processes.

1.4.1 Drone Request and Assessment (O1)

The objective is to establish a system where users can request a drone deployment through the app to the accident location. The deployed drone will record the accident scene with high-quality, real-time footage. This capability is critical because research shows that First Responder Drones (FRDs) have a shorter response time than conventional response systems [1], allowing them to bypass traffic.

1.4.2 Automated Documentation (O2)

The objective is to eliminate the NVA paper process by automating data collection. After the drone records the scene, the accident parties will receive an SMS link to fill out and submit the statement form electronically. This replacement of manual processes with automated technology is designed to speed up response time and raise the quality of accident reporting [1].

1.4.3 Enhanced Response Time (O3)

The project's quantifiable objective is to significantly enhance the overall cycle time process. The simulation results confirm that the integration of drones and automation in the Najm claim handling process optimizes the procedure, demonstrating an overall improvement rate of 40% in performance. This was achieved by reducing the total process time from 58,500 minutes (As-Is model) to 35,000 minutes (To-Be model) for 1000 instances [1].

1.5 Motivation

The strategic shift of NajmTech is guided by a four-part mission: Humanitarian Needs, National Alignment, Technological Feasibility, and the Computer Science student mission. The Humanitarian Mission seeks to make road accident procedures significantly faster and less stressful, directly enhancing the safety and well-being of motorists. This initiative aligns with the National Alignment Mission by supporting Saudi Vision 2030 goals for innovation and sustainable growth. Furthermore, the success of the To-Be simulation, which demonstrated a 40% improvement in process time, proves the technological feasibility of using drones and automation to overcome current system limitations. Finally, the Computer Science students' mission is fulfilled by applying acquired university knowledge to this critical societal need, confirming our professional competence in using innovative technology for social benefit.

2. Methodology Description

This section details the methodical approach employed for the NajmTech project, commencing with the critical phase of requirement gathering and analysis, and logically progressing through the selection of the system design, the definition of the architecture, the choice of essential technologies, and finally, the strategy for development and deployment. The methodology is specifically designed to overcome the limitations of the traditional accident response system and achieve the ambitious goals of enhanced efficiency and response time, as targeted in the project's objectives.

2.1 Requirement Gathering and Analysis

Requirement gathering and analysis constituted the foundational phase of the NajmTech project, aiming to clearly define the system's objectives, functional capabilities, and operational constraints. This phase was critical for ensuring the developed solution directly addresses the documented weaknesses of the existing manual accident response process, such as the delay caused by the "Arrive at the Destination" activity, which averages 20 minutes per instance. Requirements were systematically collected and analyzed using data derived from a fusion of real-world operational experiences and official documentation from Najm [1]. The investigation utilized qualitative analysis, such as Root-Cause Analysis (Fishbone Diagram), to pinpoint specific issues like poor image quality and paper documentation risks. Quantitative analysis, specifically flow analysis and simulation using Bizagi software, confirmed that activities classified as Non-Value Added (NVA), such as Take Pictures and Fill Statement Paper, were process bottlenecks. This diagnostic work informed the division of requirements into four categories: User, System, Functional, and Non-Functional, which collectively form the blueprint for the subsequent system design.

2.2 System Design and Architecture

The system is based on a layered (three-tier) architecture, which splits the system into three main layers: presentation, application/business logic, and data access. The design offers scalability, ease of maintenance, and modularity in case of future upgrades.

- Presentation Layer: Facilitates real-time interaction between users, drones, and external systems (Najm and Absher), reduces the delays in the processing, and enhances the response time. (O3)
- Application Layer: It will have modules that capture real-time drone deployment, location tracking, and images to guarantee fast and precise on-site analysis. (O1)
- Data Layer: It involves the automatic storage and retrieval of accident data, drone pictures, and reports, which practically means substituting the manual paperwork with automation. (O2)

The choice of this architecture was for its clear separation of concerns is provided, it can be developed in parallel, and testing and scalability in the future are made easier.

2.3 Technology Selection and Implementation

NajmTech application has a technological background based on a strategic choice of tools and platforms designed to ensure high performance, scalability, and the smooth integration of its fundamental drone and automation functionality. The system is a three-tier layered architecture, hence requiring strong technologies in every layer. The Presentation Layer (containing the mobile and web user interface) requires a modern and cross-platform framework that can provide a pleasing user experience and support real-time communication with the drone request controller. Application/Business logic layer will be based on effective server-side technologies aimed at real-time data stream processing of deployed Unmanned Aerial Vehicles (UAVs) to fulfill such tasks as location tracking and image processing, and cloud-based services for scalable computing capabilities. Lastly, the Data layer needs a secure and highly available cloud-based database solution to store sensitive user data, accident tables, and high-resolution drone images, enabling the automated documentation process (O2), which replaces the traditional paper-based systems [5]. The implementation plan is based on the incremental model that allows iterative development, testing, and deployment of these core technological components. This will reduce the risk and guarantee the project goals to respond faster (O3).

2.4 Application Development

The design of the NajmTech application follows a strict scheme of three-tier architecture that aims at ensuring scalability, maintainability, and clear definition of the responsibilities, as detailed in our system design. This approach will allow running development and a subsequent

upgrade in parallel and comply with the goals of high efficiency levels and minimal response time inherent in the project (O3).

2.4.1 The implementation of the Layered Architecture

It logically separates the application into three main layers so that the business logic will not be related to the user interface and data storage.

- **Presentation Layer (Mobile/Web App UI):** The mobile/web application Presentation Layer, herein referred to as the UI layer, is all about interaction with the user and real-time communication. Under this layer, the implementation provides the interface under which end-users can register, authenticate, and audit historical incidences, and most importantly, transact drone requests. This is achieved by the real-time interchange of services with other external systems like Najm and Absher through the Application Layer to reduce delays in processing and enhance overall system responsiveness. The major functions of the layer include user registration and authentication, an interactive map on the home page, and a form that allows drone requests.
- **Application/Business Logic Layer:** The Application/Business Logic Layer processes all the core business rules and coordinates data flow. It is the main driver of NajmTech, and it consists of multiple modules:
    - **Drone Request and Location Tracking (O1)**: This module manages the lifecycle of deployment requests, the real-time location of the accident, and the transmitting of real-time status information to the drone-control systems embedded in the Najm infrastructure.
    - **Report Handling and Automated Documentation (O2):** This element takes care of the real-time imagery and clear footage of the drone and triggers an electronic statement submission link (passed through SMS) to the parties of the accident, thus replacing the manual, non-value-added paper processes.
    - **Notification System:** Manages and sends real-time status updates (e.g., "Drone Dispatched," "Report Submitted," "Authorization Successful") to the end-user.

- **Data Layer:** The Data Layer is focused on safe and effective data storage and retrieval of all system data. It is the integration of a cloud database that will enable automatic storage of vital information, which includes:
    - Username and profile information.
    - Drone deployment records.
    - UAVs images and reports on accidents in real-time.
    - User feedback submissions.

The shift to automated and safe data storage is determinant in achieving the goal of quality and efficiency of documentation, which is in manual paperwork.

2.4.2 Integration with External Systems

The system should be interoperable with other systems, the functionality of which is not part of this project.

- **Najm System:** The direct API connection will be required to activate the drone deployment and to get the raw damage-detection and accident-location information of the drone infrastructure.
- **Absher System:** This system is necessary to register and authenticate users, perform biometric authentication, and verify the end-user information. It is well integrated and has a secure and authorized flow of processes that are in line with the national security standards.

3. Requirements

3.1 User Requirements

- **Convenience**: Users expect the app to be a convenient and efficient way to report accidents.
- **Accuracy**: Users expect the drone to accurately detect damage on the street to facilitate effective cleanup operations.
- **Timeliness**: Users expect timely responses to their requests and notifications regarding progress.
- **Feedback Mechanism:** Users should have the ability to provide feedback on the overall user experience to help improve the service.
- **Goal Tracking and Progress Visualization:** The app should allow users to track the progress of their accident reports and drone deployments, providing clear visualization of goals and milestones reached.
- **Integration with Emergency Services:** Users may expect seamless integration with emergency services for fast, coordinated responses to accidents, including automatic alerts to responders and real-time data sharing to support quick, informed decision-making.
- **Accessibility Features:** The app should incorporate accessibility features such as voice commands, screen readers, and adjustable font sizes to cater to users with disabilities and ensure equitable access to motorist services.

3.2 System Requirements

- **Compatibility**: The app should be compatible with a wide range of devices and operating systems to ensure accessibility to a broad user base.
- **Data Management**: The system should effectively manage and store user data, drone deployment records, and feedback submissions securely and efficiently.
- **Integration**: The system should seamlessly integrate with existing accident reporting systems and emergency response protocols to enhance coordination and effectiveness.
- **Performance Monitoring:** The system should include mechanisms for monitoring performance metrics, such as response times and accuracy of damage detection, to continuously optimize operations.

- **Disaster Recovery:** The system should include strong disaster-recovery measures to minimize disruption during failures or data breaches and ensure continuous service.

3.3 Functional Requirements

- **User Registration/Login**: Users shall be able to create accounts or log in to the Najm Tech app to access its features.
- **Drone Request Feature**: Users shall be able to request a drone deployment through the app to the accident location.
- **Drone Deployment**: The app shall facilitate the deployment of drones to the specified accident location.
- **Damage Detection**: The deployed drone shall be capable of detecting damage on the street caused by the accident accurately.
- **Damage Reporting:** The application shall be able to receive and accept user reports on the effects of accidents in the streets and take appropriate action.
- **Notification System**: Users shall receive notifications about the status of their request for the drone deployment process.
- **User Feedback**: Allow users to provide feedback on the effectiveness and efficiency of the cleanup process and overall experience.
- **User Logout**: Users shall be able to log out from the Najm Tech app after they finish accessing its features.

3.4 Non-Functional Requirements

- **Reliability**: The system should be highly reliable, ensuring that drone deployments and damage detection occur without failures.
- **Security**: Ensure that user data is securely stored and transmitted.
- **Performance**: The app should perform efficiently, with minimal latency in processing drone requests, and detecting damages.
- **Scalability**: The system should be scalable to accommodate a growing user base and increased drone deployment requests.
- **Usability**: The app should have an intuitive user interface, making it easy for users to request drone deployments and track the progress.

4. Architecture Design

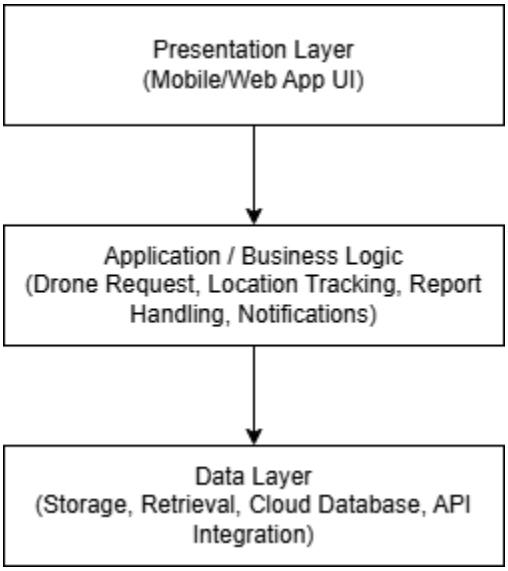

Figure 1: Architecture Design for NajmTech System

The NajmTech system will be developed based on a three-layer architecture, which was selected to provide high scalability, the separation of duties, and the possibility of the efficient processing of real-time accident data. The functions of each layer are different but function in unison with others to assist in features like deployment of drones, automated documentation, identity verification, and quick interaction with the users [5].

4.1 Presentation Layer

This layer represents the entire user interface of NajmTech, including mobile and web interfaces. It handles all direct communication with the final consumer, such as registration, login, requesting a drone, incident reporting, history view, and real-time updates of the status.

The Presentation Layer includes elements that are interactive, including maps of user positions and drone actions, incident reporting forms, and notification alerts. It has no business logic but forwards user actions to the Application Layer and awaits the outcomes.

The Presentation Layer allows easy updates by isolating interface and system logic and provides a positive user experience, particularly in critical situations like accidents, while still offering rapid, responsive interactions.

4.2 Application/Business Logic Layer

Mediator between the user interface, the database, and third-party systems (Najm and Absher). It implements all system rules and organizes the entire process of accident management.

In this tier, user requests are checked, executed, and converted into actions. Upon being requested by a user, this layer can be used to deploy a drone by communicating with the Najm system to begin deployment, retrieve the live position of the drone, and control the stream of images and status updates. It also handles automated documentation by triggering digital accident forms, processing submissions of reports, and creating notifications at each point of the process.

Additionally, it is responsible for identity verification by interacting with the Absher system to ensure that all operations are secure and authorized. This layer guarantees that the system is operating reliably, consistently, and efficiently, while at the same time ensuring that all sensitive logic is centralized and protected.

4.3 Data Layer

Data Layer offers secure and organized storage for all the data that NajmTech works with. It handles databases with user profiles, vehicle information, drone deployment history, and high-resolution images of drones.

This layer ensures the integrity, availability, and confidentiality of data, supporting backup routines, recovery mechanisms, and optimized access for high-demand operations. Since accident data may be legally important, the Data Layer should not be exposed to inaccuracies or insufficient long-term storage.

This layer enables the system to be easily scaled with an increase in the number of users, drones, and reports because it separates storage, logic, and presentation.

5. Models

5.1 Use Case Diagram

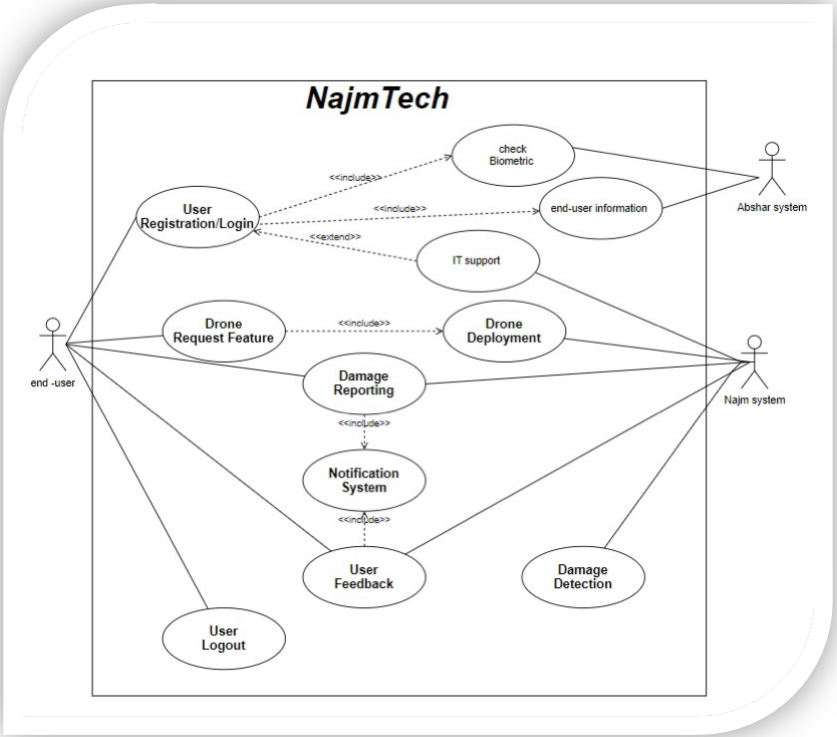

Figure 2: Use Case Diagram for NajmTech System

The stakeholders in NajmTech are the end user, the Najm system (the system of systems contains drone systems, image analysis systems, systems that maintain the drone's connection to the system, and other systems that assist the functions provided by the application), and the Absher system for security matters [6]. The end user begins by creating an account and entering biometrics (national ID, phone number, and name), which are verified through Absher and show the rest of the information, such as the cars the end user owns, plate numbers, etc., information needed by the application, and if a malfunction occurs, the application allows the end user to request technical support. The Najm system will automatically deploy the drone to the specified accident site after the end user has requested the drone to the accident site. The end user can write reports on the effects of accidents on the streets, take appropriate action, read notes on the effectiveness and efficiency of the cleaning process and the overall experience, and get notifications after each response to the reports or modification of the feedback, save the drone reports of detecting damage on the app, and log out of NajmTech.

5.2 Activity Diagram

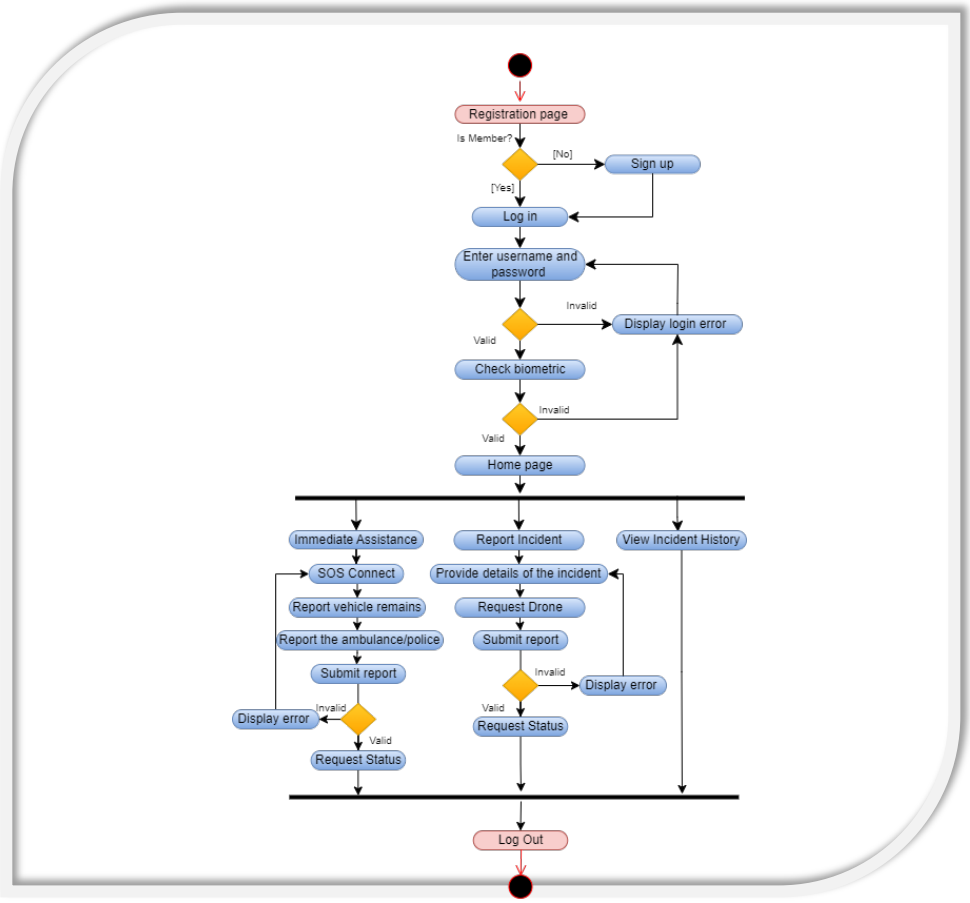

Figure 3: Activity Diagram for NajmTech System

The activity diagram illustrates comprehensive user flow, which starts with the Registration page, where the user is first made to identify if they are new or existing. Successful login requires the user to submit credentials with a biometric verification action, with failure leading to automatically redirecting the user to the login screen, and success will allow the user to access the Home page. The workflow as it is provided on the Home page directly splits into three main pathways of service: (i) Immediate Assistance, which gives an immediate SOS connection, linking the user directly to emergency services for immediate help before consulting the session. (ii) View Incident History, and (iii) Report Incident process. The pathway of reporting is the most detailed, as it requires the user to provide specific details of the incident prior to making reports on the vehicle remnants, ambulance or police needs, or the purchase of a drone. Every action of reporting is followed by a final Submit report action, followed by a check on the Request Status. Regardless of the selected path, the whole user session is finally ended with a final Logout action.

5.3 Sequence Diagram

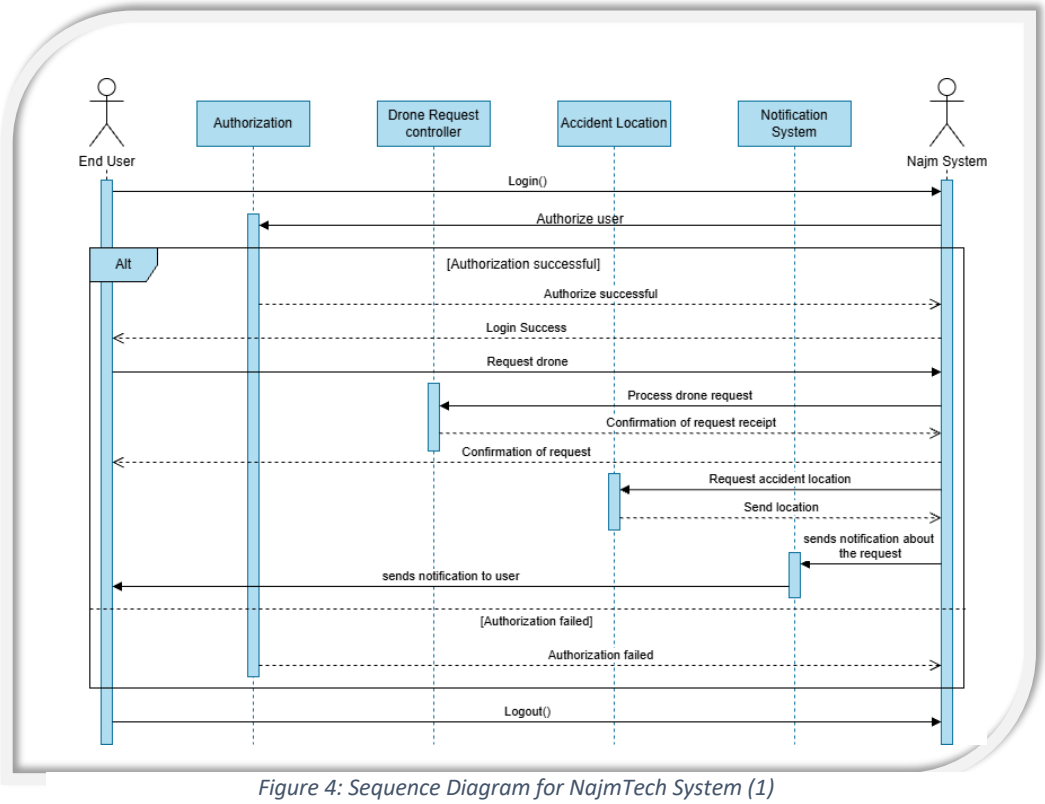

Figure 4: Sequence Diagram for NajmTech System (1)

5.3.1 Sequence Diagram Figure (1).

The above sequence diagram represents the drone request and deployment process that is used in the NajmTech application. The workflow starts as soon as a Login() is initiated by an End User and thus the authorization sequence starts. After a successful authorization, the user is authorized to send a request drone command. This command is immediately handed off to the Drone Request Controller, which takes up the task of executing it by sending back to the user confirmation of request receipt, which maintains transparency and provides instant feedback. At the same time, the system also automatically initiates the Accident Location service to obtain the exact location of the incident. The Najm System then uses the Notification System to send a real-time status report to the End User (e.g., the drone has been sent). In case the first authorization is not successful, the alternative path (ALT) ensures that the user is at once notified of the failure, thus eliminating further access to the core functions. The whole process is a model of simple, automated, and safe working flow, which reduces human delays and gives maximum response efficiency.

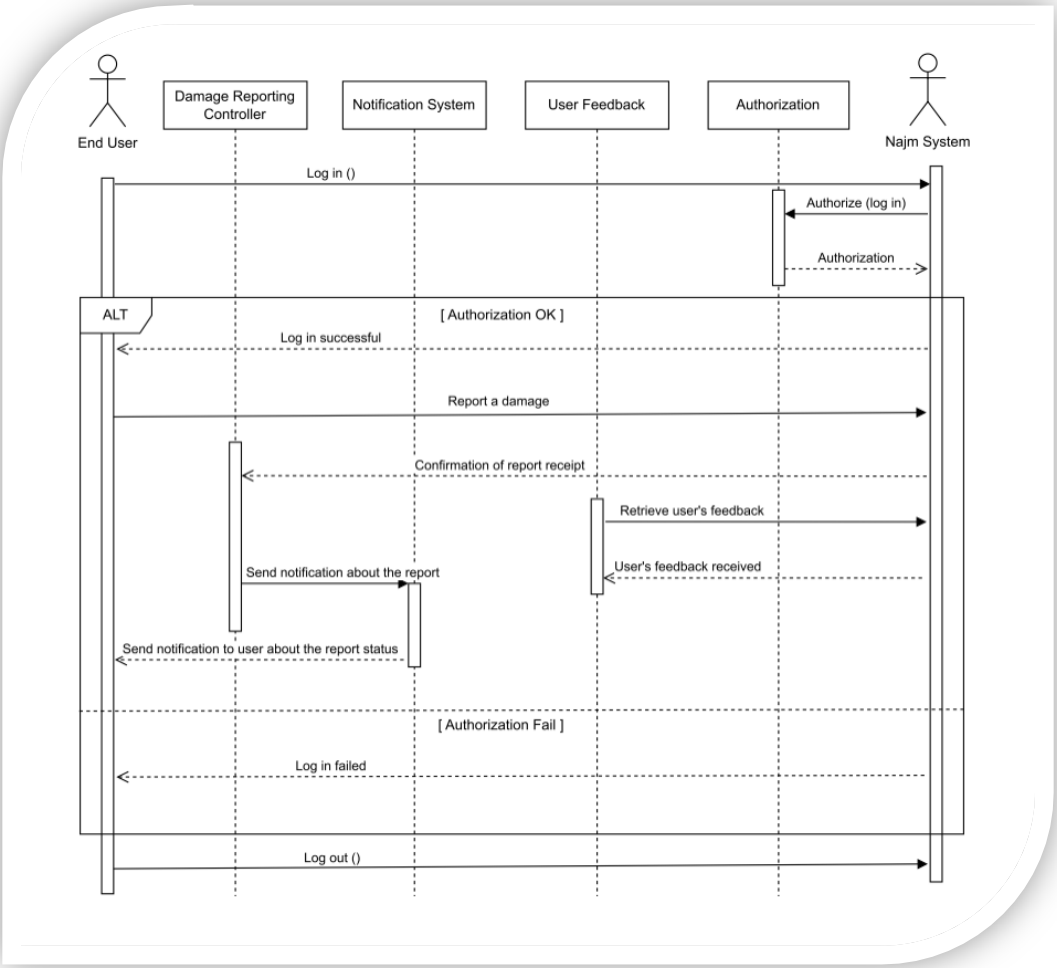

Figure 5:Sequence Diagram for NajmTech System (2)

5.3.2 Sequence Diagram Figure (2).

The sequence diagram above defines the workflow of the NajmTech System with respect to damage reporting, starting with user authentication. The workflow starts with an end user calling the Login operation, and this triggers an authorization verification done by the Najm system. The control flow splits depending on the result of this authentication: when the authorization is successful (marked with the Authorization OK), the user is notified and allowed to proceed on to the step of reporting damage. The Damage Reporting Controller takes in the report presented and then gives an acknowledgement that it has received the report and thereafter informs the Notification System. Subsequently, the User Feedback component retrieves and receives the user's feedback from the Najm System. Concurrently, the Notification System sends an update on the report's status to the End User. If authorization is unsuccessful ("Authorization Fail"), the system immediately notifies the user that the Login failed. The overall interaction concludes with the user performing a Logout().

5.4 Context Diagram

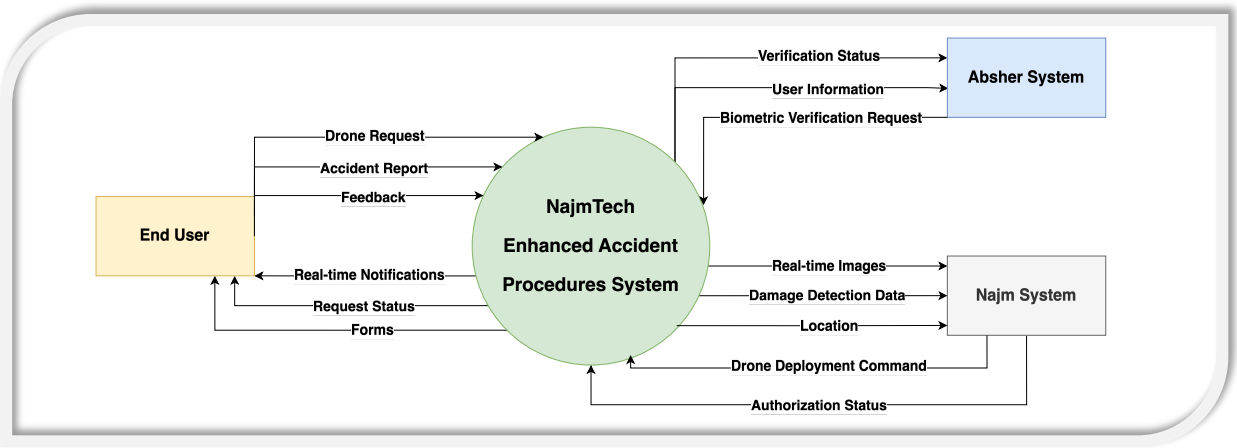

Figure 6: Context Diagram for NajmTech System

The Context Diagram establishes the high-level operational statement of the NajmTech Enhanced Accident Procedures System, confirming the fact that it is a single, central process that can control the entire accident lifecycle-user verification to operational deployment of the drones and the final reporting.

The system is connected to three key external entities:

- **End User:** The End User is the input provider that provides the Drone Request and Accident Report and receives updates on services, including real-time notifications and digital forms.

- **Absher System:** It is a body that handles security and verification. NajmTech begins with the process of sending the Biometric Verification Request and, in turn, absorbs the Verification Status and authorized User Information, which is sent by Absher [6].

- **Najm System:** This is an operational partner to whom NajmTech provides the Drone Deployment Command and Location. As a payback, Najm will offer essential operational information, such as real-time images, damage detection data, and authorization status.

6. Prototype

6.1 Splash Screen

The splash screen displays the NajmTech logo prominently at the center of a clean, minimal background. Its purpose is to introduce the system's identity, ensure smooth visual transition during application loading, and set a professional tone before the user enters the onboarding flow.

Figure 7: Splash Screen

6.2 Onboarding Screen I – "Anywhere You Are"

This onboarding screen introduces the system's capability to assist in any location. It features an illustrative graphic representing remote accident support via drones, accompanied by a concise descriptive caption. A central action button allows progression to the next screen, while a "Skip" option in the header enables bypassing the onboarding sequence.

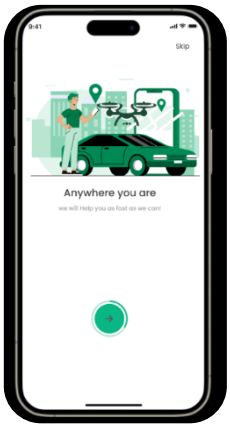

Figure 8: Onboarding Screen I – "Anywhere You Are"

6.3 Onboarding Screen II – “At Anytime”

The second onboarding screen highlights the 24/7 availability of NajmTech services. A visual illustration reinforces the concept of continuous support for motorists. Consistent navigation elements such as the bottom action button and top “Skip” link maintain clarity and ease of use throughout the onboarding experience.

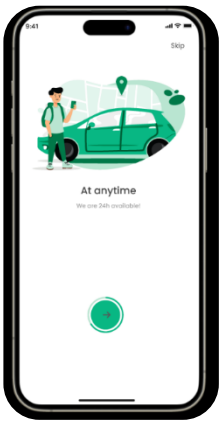

Figure 9: Onboarding Screen II – “At Anytime”

6.4 Onboarding Screen III – “Try It Now”

The final onboarding screen encourages users to begin interacting with the application. The illustration emphasizes real-time tracking and location-based service delivery. A confirmation button labeled “Go” leads users to the main authentication interface, providing a smooth transition into the system.

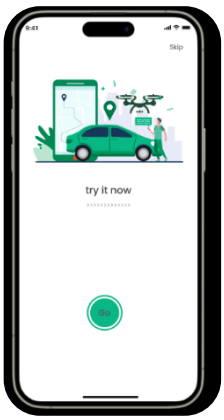

Figure 10: Onboarding Screen III – “Try It Now”

6.5 Welcome Screen

This screen greets users with a welcoming visual and message summarizing the purpose of the application. It provides two primary actions: Create an Account for new users and Log In for returning users. The design ensures clarity, simplicity, and ease of navigation into the authentication process.

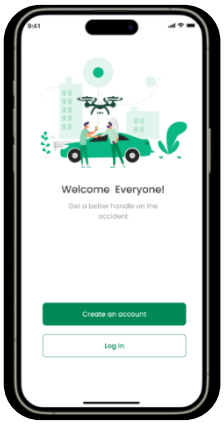

Figure 11: Welcome Screen

6.6 Registration Screen

The registration screen enables new users to create an account using their email or mobile number. It includes structured input fields for name, email, phone number, and gender. A mandatory agreement to the Terms of Service ensures compliance with system policies. Social authentication options (Google, Facebook, Apple) are provided to simplify onboarding and improve accessibility.

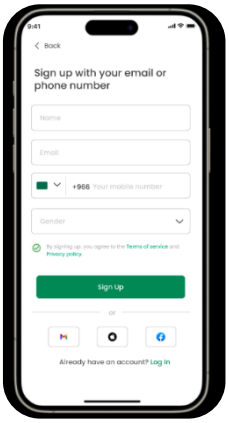

Figure 12: Registration Screen

6.7 Login Screen

The login screen allows authenticated users to access the system by entering their email or phone number along with their password. It includes a “Forgot password?” feature for credential recovery and offers alternative login methods through integrated social platforms. A link to the registration screen is provided for users without existing accounts.

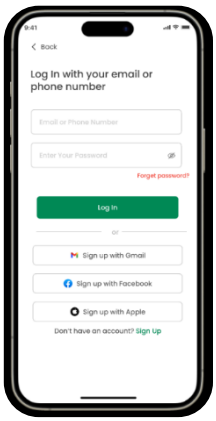

Figure 13: Login Screen

6.8 Phone Verification Screen

This screen conducts identity verification through a One-Time Password (OTP) sent to the user’s registered phone number. The interface includes six dedicated input slots for the verification code and an option to resend the OTP if needed. A “Verify” button completes the authentication process and transitions the user into the main application environment.

Figure 14: Phone Verification Screen

6.9 Home & Drone Map Interface

The Home Screen serves as the central operational interface of the NajmTech application. It features an interactive map displaying the user's current location alongside nearby active drones, visualized through animated drone markers and detection radii. A search bar enables users to locate the nearest available drone or relevant service points quickly. The bottom navigation bar provides seamless access to essential modules, including History, Accident Reporting, User Profile, and Settings, ensuring efficient movement across the system. This interface is designed to support real-time situational awareness and facilitate immediate access to accident-related services.

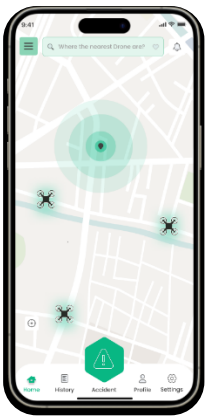

Figure 15: Home & Drone Map Interface

6.10 Slide-Out Profile Menu

The Slide-Out Profile Menu provides users with quick access to personal and system management options. The top section displays the user's profile image, name, and email address, reinforcing identity confirmation. Beneath this header, the menu presents structured navigation items, including History, Complaint Submission, About Us, Settings, and Help & Support. A dedicated Logout option ensures secure session termination.

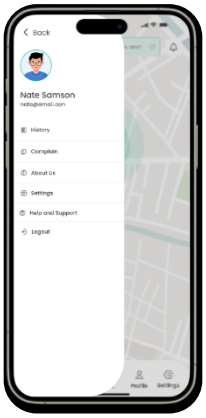

Figure 16: Slide-Out Profile Menu

6.11 Transport Selection Screen

The first screen in the reporting flow requires the user to select their transport type. This page presents four distinct options: Car, Bike, Cycle, and Taxi, visualized with clear icons, which are likely essential details for processing the incident report and dispatching the correct resources. This streamlined selection ensures accurate documentation begins immediately.

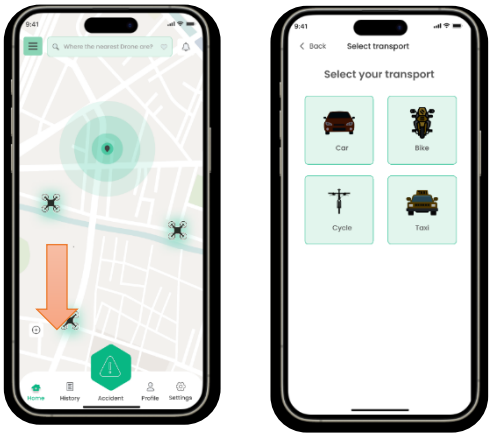

Figure 17: Transport Selection Screen

6.12 Drone Deployment and Tracking Screen

Upon submitting the request, the user is immediately taken to a real-time tracking map showing the drone's status. The interface displays the path of the deployed drone from its origin to the accident scene and provides an estimated arrival time (e.g., "Your drone is coming in 3:35"). This page also prominently features the assigned Najm Tech team members (e.g., "Khalid") and provides essential real-time communication options: a "Call" button for immediate voice contact and a "Message" button to initiate a chat.

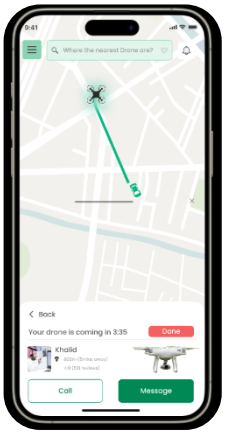

Figure 18: Drone Deployment and Tracking Screen

6.13 Call Interface

The application includes a dedicated, full-screen call interface for users who prefer direct voice communication with the Najm Tech team. This screen displays the profile of the contact (e.g., "Khalid Najm") and features standard telephony controls, ensuring rapid, direct access to professional assistance during a high-stress accident scenario.

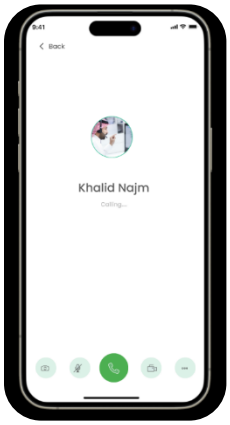

Figure 19: Call Interface

6.14 Chat Interface

The chat interface facilitates instant and essential communication between the user and the assigned Najm Tech team member. The chat history shows the initiation of the report (e.g., "I had an accident") and the system's response, which typically involves confirmation and a critical triage question, such as inquiring about the need for an ambulance. This feature minimizes verbal miscommunication and provides a documented record of the initial incident assessment.

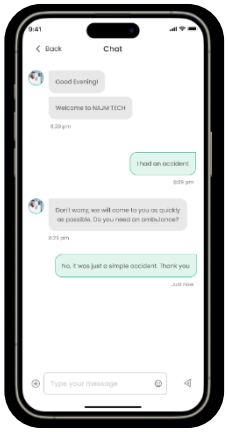

Figure 20: Chat Interface

6.15 Service Confirmation and Feedback Screen

Once the incident is resolved, the user is presented with a confirmation screen displaying a "Thank you" message and a successful submission checkmark. This is immediately followed by the user feedback mechanism, which prompts the user to rate the service, typically on a five-star scale, and offers a text box for providing detailed feedback. This step is vital for the system's continuous improvement, aligning with the user requirement for a feedback mechanism.

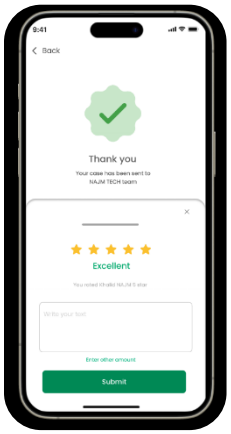

Figure 21: Service Confirmation and Feedback Screen

6.16 History Page

The History page provides a log of past interactions and reports submitted by the user. It is designed to offer a record of relevant activities, such as past drone deployments and incident reports. Each entry in the history list is clearly marked with a location identifier (e.g., "Office," "Home," "House") and a specific address, allowing the user to review their history of assistance requests easily. This feature fulfills the requirement for goal tracking and progress visualization by allowing users to audit their interactions with the system.

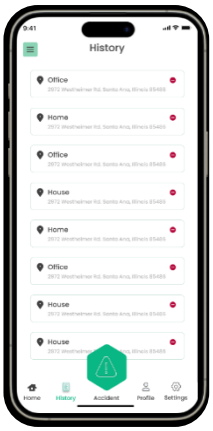

Figure 22: History Page

6.17 User Profile Page

The User Profile page displays personalized information about the user. The design is structured to allow users to view and potentially edit their account details. Key fields visible include the user's name (e.g., "Mohamed"), email address (e.g., "nate@email.com"), mobile number, gender, and address. Critically, a prominent logout button is included to ensure secure session termination. This page serves as the central hub for account management and identity confirmation.

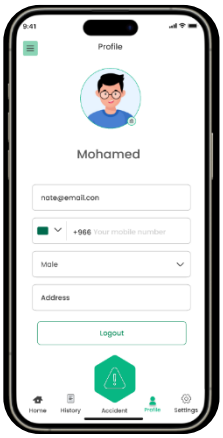

Figure 23: User Profile Page

7. Ghant Chart

The following table and visualization represent the project schedule and individual responsibilities for the final report sections.

ID	Task Name	Start	End	Duration	Resources
1	Introduction (Project Description)	2025-11-10	2025-11-11	2 days	Hissah Almuhaysh
2	Methodology	2025-11-12	2025-11-13	2 days	Team
3	Req. Gathering & Analysis	2025-11-12	2025-11-13	2 days	Hissah Almuhaysh
4	System Design & Architecture	2025-11-12	2025-11-13	2 days	Ftoon Althafar
5	Technology Selection & Implementation	2025-11-12	2025-11-13	2 days	Rana Althafar
6	Web/Desktop Application Development	2025-11-12	2025-11-13	2 days	Latifah Alhafith
7	Requirements (User, System, Functional, Non…	2025-11-16	2025-11-18	3 days	Team
8	Architecture Design	2025-11-19	2025-11-20	2 days	Ftoon Althafar
9	Models	2025-11-23	2025-11-24	2 days	Team
10	Use Case Diagrams & Descriptions	2025-11-23	2025-11-24	2 days	Ftoon Althafar
11	Sequence & Activity Diagrams & Descriptions	2025-11-23	2025-11-24	2 days	Rana Althafar
12	Context Diagram & Description	2025-11-23	2025-11-24	2 days	Latifah Alhafith
13	Prototype Generation & Description	2025-11-25	2025-11-27	3 days	Team
14	Final Review, Formatting, and Submission	2025-11-30	2025-11-30	1 day	Team

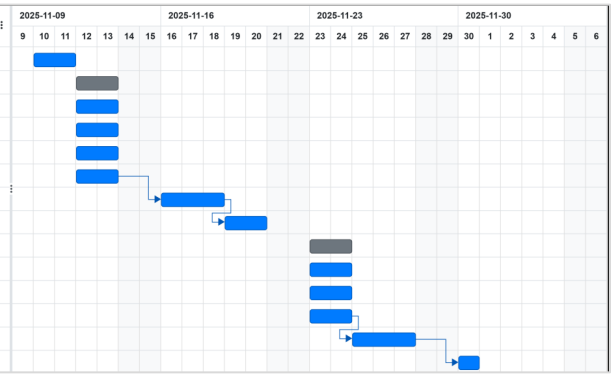

Figure 24: Work Timeline

8. Conclusion

In summary, the NajmTech system is a major development in infrastructure management because of its sophisticated drone requests, drone deployment, damage reporting, and damage detection capabilities. The system's use of drones allows it to quickly and precisely detect and report road damage, enabling prompt maintenance and repairs. By being proactive, this reduces the possibility of traffic jams and improves road safety. The effective use of resources and quick incident response made possible by the smooth integration of these operations have a revolutionary effect on infrastructure dependability and urban mobility. All things considered, the NajmTech system is a prime example of how cutting-edge technology can enhance everyday living in urban settings and optimize infrastructure management.

References

[1] Elhag, S. M., Shaheen, G. H., & Alahmadi, F. H. (2023). Accident response time enhancement using drones: A case study in Najm for Insurance Services. I.J. Information Technology and Computer Science, 15(6), 1–14. https://doi.org/10.5815/ijitcs.2023.06.01

[2] Najm Insurance Services, "Najm," [Online]. Available: https://najm.sa/home

[3] Gohari, A., Ahmad, A. B., Rahim, R. B. A., Elamin, N. I. M., Gismalla, M. S. M., Oluwatosin, O. O., Hasan, R., Latip, A. S. A., & Lawal, A. (2023). Drones for road accident management: A systematic review. IEEE Access, 11, 111253–111275. https://doi.org/10.1109/ACCESS.2023.3321142

[4] A. Gohari et al., "Drones for Road Accident Management: A Systematic Review," in IEEE Access, vol. 11, pp. 109247-109256, 2023, doi: 10.1109/ACCESS.2023.3321142.

[5] I. Sommerville, *Software Engineering*, 10th ed. Boston, MA, USA: Pearson Education Limited, 2015, ISBN: 978-0133943030.

[6] "Registration and Subscription," Absher Individuals, Ministry of Interior – Kingdom of Saudi Arabia, Feb. 13 2024. [Online].